

Software Design Principles

Discussion 08 – CSC 4033

Readings

These reading selections from Software Engineering (Sommerville) aren't quite as lengthy as the page numbers suggest, as they are full of diagrams and images. We're not so concerned with applying the Unified Modeling Language (UML) used in these, so focus more on the concepts being discussed and use the UML diagrams to support your learning.

Required

- 1) Software Engineering, Ch.5 Intro-5.3 (15 pages)
- 2) Software Engineering, Ch.6 Intro & 6.3 (12 pages)

Recommended

- 1) Be on the lookout for links to additional web-only sections of the book
- 2) Any resources on software design paradigms (layered, data-oriented, object-oriented, event driven)
- 3) Any resources discussing software architectures
- 4) "Programming-in-the-large versus programming-in-the-small", paper by Frank DeRemer and Hans Kron

Learning Outcomes

- Discuss the role of system models in the design process
- Explain how context, interaction, and structural models support the design process and what kinds of boundaries, dependencies, and relationships each identifies
- Compare the concept of "architecture in the large" to "architecture in the small"
- List at least five different architecture patterns and briefly describe the key ideas and concepts for each